National College of Ireland

Project Submission Sheet

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Programme:	Msc in Cloud Computing	**Year:**	2025-2026
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Lecturer:			
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Signature:	Meenakshi
Date:	8 july 2026

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Your Name/Student Number	Course	Date

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AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

Description of AI Usage

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[Insert Tool Name]	
[Insert Description of use]	
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Additional Evidence:

[Place evidence here]

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Governance, Compliance and Ethics Evaluation of Terraform Automation vs manual provisioning for Cloud Infrastructure

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***Abstract*—Cloud computing has become the backbone of the IT landscape, providing organisations with scalable and reliable applications. This report discusses the governance, compliance and ethical considerations of a terraform-based research project that compares Infrastructure as Code (IaC) provisioning of cloud infrastructure against manual provisioning on Amazon Web Services (AWS). This project proved that Terraform maintained deployment consistency, reduced provisioning time, and reduced manual configuration errors. On the basis of these results, this report critically assesses the ethical and regulatory issues raised by cloud automation across the system lifecycle from design to deployment and operation, and after deployment. Specific focus is on governance principles including transparency, security, privacy and accountability as well as compliance with General Data Protection Regulation (GDPR), Irish Data Protection Act 2018 and ACM Code of Ethics. The report tells that the use of the Terraform are consistency and operational efficiency but that responsible cloud automation also means effective governance, human over sight and ongoing compliance monitoring.**

I. Introduction

Cloud computing has changed the way organizations deploy, manage and scale applications, offering flexible, on-demand computing resources. With the rise in the cloud's popularity, organizations need a dependable way to provision infrastructure and ensure that it remains secure, consistent and compliant with regulations. This report is based on a research project that looked at the differences between a standard manual approach to cloud infrastructure provisioning and Infrastructure as Code (IaC) in this instance, with Terraform running on Amazon Web Services . The project tested and measured both methods for deployment time, deployment consistency, accuracy of deployment configuration and operational efficiency. The experimental results showed that Terraform dramatically cut down deployment time, delivered repeatability and minimized human mistakes in infrastructure deployment.

While the project was primarily about deployment consistency and not customer information, there are important governance, compliance and ethical considerations involved in managing cloud infrastructure. Certain elements of infrastructure configuration, system logs, identity and access management (IAM) policies, or deployment records might contain information that needs to be appropriately protected. Security risks, data breaches, service outages, and compliance issues are potential risks that can arise in poorly managed cloud environments. Therefore, cloud engineers must take an ethical stand in building, operating and maintaining infrastructure in a way that is secure, transparent and accessible.

There are multiple legal and professional standards to help create a responsible cloud infrastructure. Cloud environments involve the processing of personal data and the General Data Protection Regulation (GDPR) provides for principles, including data minimisation, integrity, confidentiality, accountability and security. In Ireland, these requirements are backed up by the Data Protection Act 2018. While the project's experimental setting did not specifically deal with personal information of customers, cloud platforms often include administrative information like authentication records, audit logs, IP addresses and user accounts, which can be subject to the GDPR.

Accordingly, governance and ethical issues must be embedded into the design, deployment, implementation and maintenance of cloud infrastructure in order to maximize the efficiency of cloud automation while maintaining security, accountability and public trust.

II. Evaluation

A. *Design and Architecture*

Cloud infrastructure, its design and architecture are fundamental to ensuring that systems are secure, reliable, compliant and ethically managed. The same AWS infrastructure was provisioned in the research project and through AWS Management Console and through automated provisioning with Terraform. While the main goal was to focus on deployment consistency and efficiency, there are also governance and compliance considerations to take into account for the architecture options. With Terraform's Infrastructure as Code (IaC) approach, infrastructure can be described in code, allowing deployments to be repeatable, version controlled and auditable unlike manual infrastructure provisioning.

Changes to manual infrastructure may be hard to track, as the cloud console can see to administrators making changes that haven't been documented. This means that it is harder to maintain consistency between the development, test and

production environments, which can lead to configuration drift. Terraform, on the other hand, represents infrastructure as code. Code that can be stored in a version control system like Git that is then applied to build infrastructure.

The compliance to legal and organizational policies should also be included in the architectural design. The infrastructure should therefore be designed based on the principles of General Data Protection Regulation (GDPR) such as minimisation, confidentiality, integrity and accountability of data.

Ethically speaking, Terraform is good as making infrastructure configurations open to review before deployment. The terraform plan command gives engineers visibility into all the changes the they are about to make in the infrastructure, allowing them to identify any problems before they impact their production environments. But, automation isn't a way to show moral obligation

TABLE I
KEY GOVERNANCE AND COMPLIANCE CONTROLS

Area	Control	Purpose
Governance	Git Version Control	Tracks changes and supports accountability.
Access Control	AWS IAM & MFA	Restricts access to authorised users.
Security	Security Groups	Protects cloud resources.
Data Protection	Encrypted S3 State	Secures Terraform state files.
Compliance	GDPR & DPA 2018	Supports legal compliance.
Monitoring	CloudTrail & CloudWatch	Enables auditing and monitoring.

B. *Implementation and Operation*

The implementation phase and the operation of a cloud infrastructure project are very important as they will determine the security, ethical and reliable deployment and operation of the system designed. Automation isn't enough to ensure ethical or compliant results; Terraform automates provisioning of infrastructure. Organisations need to have the right governance structures in place across the development, deployment and running of an organisation.

A major ethical aspect that has to be considered during the implementation is accountability. While Terraform automates the deployment of infrastructure, it is up to the engineers and administrators who write, review and approve the infrastructure code to make decisions about deployment. Automating should, therefore human decision making. In professional DevOps environments, the Terraform configurations should be version-controlled and be subject to pull requests for changes prior to being merged into production environments. This will give clarity over who has made the changes to the infrastructure and help organisations know who has made a specific change if there is a security incident or some sort of operational failure.

Another key governance needs is transparency. One of the benefits of Terraform is that any infrastructure changes can be seen prior to deployment via the terraform plan command. This enables engineers to review changes they are proposing to make to the resource, see if these proposed changes would cause the creation of resources that they didn't intend, or the deletion of resources they didn't intend to delete, and ensure that they are in line with any organizational policy changes. Reducing operational risks and supporting regulatory audits is another advantage of transparent deployment processes, as it ensures that all infrastructure changes can be documented, therefore boosting organizational trust.

Monitoring and compliance testing should be continuous and a part of operational governance. Using automated security scanning tools that can detect insecure network configurations, too many permissions, public storage services, or outdated software, infrastructure should be regularly reviewed.

C. *Post-Deployment Implications*

The ethics and governance obligations for cloud infrastructure don't end once it's up. While Terraform enhances deployment uniformity and operational effectiveness, organisations are required to make sure they remain secure, compliant, as well as sustainable throughout their lifespan. To ensure operational risks are kept to a minimum, and trust remains in automated infrastructure management, regular monitoring, auditing and maintenance activities are critical.

Changes to infrastructure configurations are recorded as code in Terraform, making it easy to use version control systems like Git to keep track of changes. This will give a full account of any infrastructure changes, enabling organisations to trace back security incidents, prove compliance with regulations and to aid any internal or external audit. Infrastructure as Code is a huge boost in accountability and governance with no un documented changes that can happen when done using manual provisioning.

Another key factor is sustainability of the environment. Data centers consume a lot of energy when they are running through the cloud. If these resources are not managed properly, they can lead to virtual machines, storage volumes, or networking resources that are not used and wasting computing resources. By automating the process of creating and destroying infrastructure, Terraform can help organizations to reduce the waste of resources and to make their cloud operations more sustainable. But deployed resources need to be reviewed by the engineers on a frequent basis to prevent such costs and environmental effects.

Lastly, it is necessary to keep human control. Rather than replacing the professional judgement, automation should be used to help manage the infrastructure. Cloud administrators must be able to review infrastructure changes, approve production changes, make manual changes during incidents and rollback the changes when production changes do not work out. Automation and effective governance ensures that cloud infrastructure is safely and securely operated, transparent, accountable and ethically governed across its lifecycle.

III. CONCLUSION

This report is a critical analysis of the governance, compliance and ethics of a cloud infrastructure project focusing

on the provisioning of cloud infrastructure using manual processes versus Terraform automation. This analysis reveals that Terraform not only increased deployment speeds, but it also increased deployment consistency and operational efficiency which are great benefits but responsible cloud automation has gone beyond just that. Securing the design of the system, access control, deployment methods, monitoring the system in real time, and accountability for all parts of the infrastructure should be elements of effective governance. Following legal guidelines like the General Data Protection Regulation (GDPR) and the Irish Data Protection Act 2018, along with the ACM Code of Ethics, ensures the secure and responsible use of cloud infrastructure. While Automation as Code is a great way to minimise human error and configuration drift, it is important to always have the right human oversight, security reviews, and organisational governance policies in place.

[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]

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